



Essay 2: Build a 15-20 Minute Guided Learning Activity

Due Week 09: Mon, 12 Oct 2026 23:59 SGT

Submission: one Markdown file for a tutorial question, or one .ipynb file for a programming exercise.

Essay 2 contributes **10%** to the course assessment.

Build a reusable learning activity for a future offering of CP4285. You have been allocated **two possible weeks**: one from Weeks 01-05 and one from Weeks 06, 08-12. Choose **one** of those two weeks and create an activity that helps a future student reinforce a concrete example or plausible concept from that lecture.

Your submitted activity must take a student about **15-20 minutes** to complete in a guided form. It is one question or exercise within the usual format of a **40-minute tutorial or lab**, which typically contains **2-3 (multi-part) questions** in total. It may be either:

Activity type	Submission format	What it should do
Tutorial question	One Markdown (<code>.md</code>) file	Guide the student through a short worked example, then ask them to apply, extend, compare, or diagnose it.
Programming exercise	One Jupyter notebook (<code>.ipynb</code>) file	Give the student a small, runnable task that applies a lecture example, with clear setup, instructions, and expected behaviour.

The objective is not to cover an entire week or reproduce a slide deck, but make one particular idea or concept easier to practise, test, or reason about. Submit one guided question or exercise, not a complete tutorial or lab. The activity should occupy 15-20 minutes of a typical 40-minute session.

You may choose either of your allocated weeks. One is from the first half of the course and the other is from the second half; either may be a week that has already been taught or one that has not yet run.

- For a taught week, select a specific lecture example that merits rehearsal or extension.
- For a future week, use the relevant week description in the [course schedule](#) - especially its topics and learning outcomes - as the starting point for your synthesis. Create an exercise that is faithful to that planned scope without assuming access to future slides.

- State the selected week, the concept being reinforced, and the intended learning outcome near the beginning of the submitted file.

You do **not** need to use a card-suit lens or the week's ethics thread. You may use either one when it genuinely improves the exercise's motivation, scenario, constraint, or reflection question.

1 Week Allocation

Choose **one** of the two weeks assigned to your student identifier. The allocation provides a balanced spread across the course while preserving a choice between the two halves.

Student allocation: choose one week	Student allocation: choose one week
067A - Week 02 or Week 09	067Z - Week 04 or Week 08
110D - Week 03 or Week 12	111A - Week 01 or Week 06
117S - Week 03 or Week 09	123Z - Week 04 or Week 10
135E - Week 05 or Week 06	247C - Week 05 or Week 12
258H - Week 03 or Week 06	396U - Week 05 or Week 09
512A - Week 04 or Week 11	572B - Week 05 or Week 10
597J - Week 02 or Week 06	610J - Week 02 or Week 11
620L - Week 04 or Week 08	684R - Week 01 or Week 06
715E - Week 03 or Week 08	877B - Week 04 or Week 09
880X - Week 01 or Week 11	905Z - Week 02 or Week 08
908J - Week 05 or Week 11	910X - Week 01 or Week 08
940A - Week 01 or Week 12	951H - Week 01 or Week 10
963U - Week 02 or Week 12	978E - Week 03 or Week 10

2 Required Submission Structure

Design the submitted file for a future student who will accomplish the activity independently. It should contain the following elements:

1. **Title and learning goal.** Name the selected week and explain, in plain language, what the student should be able to do by the end.

2. **Brief setup or recap.** Introduce only the notation, data, code, or context needed for the exercise. Link the activity to a recognisable lecture example rather than retelling the whole lecture. You can refer to an example by concept, not (necessarily) the exact example given in class.
3. **Student task.** State the prompt, expected output, estimated time, and what the student should calculate, explain, inspect, or implement.
4. **Guidance and feedback.** Provide a solution, worked reasoning, expected output, tests, or graded hints that allow the student to check their understanding after attempting the task. This would be provided after the solution is created by the student as practice, notionally without AI use. Any worked solution should be positioned after the activity, and clearly demarcated.
5. **Attribution.** Credit any external data, code, figures, or sources used in the submitted file. Course slides and the course schedule need not be cited as external sources.

For a tutorial question, submit a self-contained Markdown (`.md`) file. For a programming exercise, submit a Jupyter notebook (`.ipynb`) that runs from top to bottom in a standard Jupyter environment, where students can fill in code blocks or provided fix buggy code. Keep data and dependencies lightweight enough that a future student can complete the activity within the stated timebox.

2.1 Course Materials, Evidence, and Citation

If appropriate, use **APA author-date in-text citations** in the activity - for example, (*Gomez-Uribe & Hunt, 2015*) or *Gomez-Uribe and Hunt (2015) argue that* End the submitted file with an APA-formatted reference list containing the sources actually cited, if citations are used. Do not treat citations as decorative background: use each source to support a specific factual or technical claim. Credit external code, data, and figures clearly.

Unlike [e1](#), if you use any of the suggested e-book readings, you must cite which e-book and chapter. You need not cite the course materials; these are implied in your assignment.

3 Assessment and Peer Review

A strong activity has one coherent learning purpose, uses a technically sound example, and gives students enough information to make progress without giving away the answer before they try. It makes the intended action and the expected result visible.

Assessment will focus on the activity’s learning design, technical quality, connection to the selected week, usability for a future student, and demonstrated ownership of the activity’s ideas and implementation. Clear communication and transparent, student-directed AI use are expected. Your activity will be assessed using the following criteria:

Criterion	Outcome
Learning design	A focused, realistic 15-20 minute guided task with a clear learning goal and an appropriate level of guidance.
Technical quality	Correct recommender systems reasoning, code, calculations, and expected outputs.
Connection to the course	A specific, meaningful connection to the selected week’s example, topics, and learning outcomes.
Submission usability	Clear sequencing, readable explanations, reproducible code where relevant, and a credible feedback or solution path.
Ownership and attribution	Student-led design decisions, transparent use of assistance, and appropriate acknowledgement of reused material.
Interestingness (Peer review only)	How interesting the activity is and its connectivity to the lecture week’s goals.

Essay 2 contributes **10%** to the course assessment, as per the same division as [e1](#), where the comment weight is designed to be heavy

Component	Weight	Arrangement
Your submitted Essay 2 activity	8%	Assessed on the quality of your individual guided learning activity.
Peer review	2%	Assess the activity allocated to you by the teaching team, using the provided peer-review process. The peer review will focus only on the Interestingness criterion of the exercise, which the teaching team will not assess (but moderate).

The teaching team will **send each student an allocated activity directly** for peer review. Complete the peer review independently. Do not use generative AI or automated analysis/writing tools to read, interpret, evaluate, draft, revise, or submit the review. Use of approved accessibility support remains subject to the course and University policy. Peer review will be facilitated through the **Canvas Assignment Component**. The peer-review period and deadline will be confirmed there.

Unlike **e1**, you are allowed to post, upload, share, or otherwise disclose your own **e2** activity on the public web or through other public-facing channels. This permission does **not** apply to an activity allocated to you for peer review, peer-review comments, or material that identifies the allocated activity's author. Do **not** post, upload, share, or otherwise disclose those peer-assessment materials through public-facing channels.

3.1 Submission Metadata, Reuse, and Credit

Place the following **Submission Metadata** at the start of your submitted file, immediately **before** your AI-use declaration:

- Your student identifier.
- The allocated week you chose and the lecture example it reinforces.
- A reuse preference: **May reuse with named credit** or **Do not reuse beyond assessment**.

The teaching team may reuse a resource in a future CP4285 offering when its author has not opted out. Whenever a submitted resource is used, the student's name will be remapped to the resource and credited for it. If you do not want your resource reused, select **Do not reuse beyond assessment** in the Submission Metadata; this preference does not affect assessment.

3.2 AI Use, Peer Collaboration, Ownership, and Verification

(This subsection is identical to **e1**'s policy)

AI use is encouraged. Use it as a research and thinking partner to identify vocabulary, locate possible lines of inquiry, explain unfamiliar recommender concepts, generate alternative activity designs, test counterarguments, debug small pieces of code, and improve the clarity of your draft. It is not evidence, however, and all AI-generated claims and citations must be verified against the original sources.

You must **lead the investigation**. Before using AI substantively, establish the scaffolding for its work: identify the selected week and lecture example, set a provisional learning goal, state the technical issue you wish to investigate, and give the system a focused task. Your prompts should steer the inquiry rather than ask AI to produce an unexamined activity. The central idea, learning design, technical judgment, organisation, and final activity must be your own. AI may

inform your thinking, but it must not replace the ideation that gives the activity its intellectual direction.

In an **Acknowledgements** section after the Reference List, attach an **AI-use declaration**. It must identify the tool(s) used; explain how you framed, constrained, or scaffolded the AI's investigation; state how AI contributed; and give two examples of outputs that you checked, corrected, or rejected. Retain your relevant prompts and outputs in case they are requested, but you need not submit the specific prompt.

Students are encouraged to discuss general aspects of their allocated course topics to further their understanding of the course's concepts. Discussion may clarify concepts and test reasoning, but students must not co-author submitted files or submit materially shared activity designs. Discussions that substantially affected your learning-design or technical decisions should also be declared in your **Acknowledgements** section. Note: the permission to discuss general course concepts does not apply to an allocated peer activity or to the preparation of your peer review.

Students may be asked to discuss or explain their activity, its sources, its pedagogical and technical choices, and their use of AI. This is an opportunity to demonstrate ownership of the work. The instructor's academic judgment, including the determination of authorship, the assessment of the work, and the final grade, is final.

4 Before You Submit

- ☐ The submitted file names one selected week and one clear learning goal.
- ☐ The activity is realistically completable in 15-20 minutes in guided form, as one of 2-3 questions in a 40-minute tutorial or lab.
- ☐ The task reinforces a specific lecture example rather than summarising a whole week.
- ☐ A student can check their work through a solution, hints, tests, expected output, or explained reasoning.
- ☐ The submission format is correct: `.md` for a tutorial question or `.ipynb` for a programming exercise.
- ☐ Code cells run in order, where programming is involved.
- ☐ Submission Metadata appears before the AI-use declaration and records the reuse preference.
- ☐ External material and AI use are acknowledged.